

Wardropian Cycles: fair and optimal traffic assignment concept for compliant connected autonomous vehicles.

Michał Hoffmann, Michał Bujak, Grzegorz Jamróz, Rafał Kucharski

Rafał Kucharski

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Jagiellonian University

1. Concept: System Optimal User Equilibrium
2. Formalisation
3. Wardropian Cycles
4. Results Barcelona + ...
5. Generalisations and applicability
6. Conclusions

Slides made by *GenZ* delivered by millenial... see how it goes.



We're all humans...

We're

egoistic



We're all humans...

We're

utility maximizers



We're all humans...

We're

egoistic utility maximizers

...at least in routing.



Wardrop (User) Equilibrium

- fair – equal travel times
- inefficient
- naturally emerging

System Optimum

- travel times differ
- efficient
- requiring external incentive

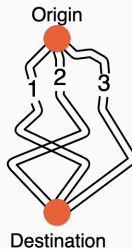
Price of Anarchy



Why to bother? - quick example

Routes for this OD in UE		
route:	flow:	time:
1	10x 	14 min 
2	30x 	14 min 
3	50x 	14 min 

$\hat{t} = 14$ min, times are equal



Routes for this OD in SO		
route:	flow:	time:
1	20x 	15 min 
2	30x 	14 min 
3	40x 	9 min 

$\hat{t} = 12$ min, times differ

What if we could combine them?



Speaking of equity while allowing unequal routes is pointless...



Speaking of equity while allowing unequal routes is pointless...

unless we change the perspective



Speaking of equity while allowing unequal routes is pointless...

unless we change the perspective

What if we broaden our point of view to a multiple-day period?



That's how the idea of Wardropian Cycles was born



Wardropian Cycles

Idea that guarantees equity for any given assignment, not instantly, but after some time.

But there are several questions:

- what is “equity” now



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Wardropian Cycles

Idea that guarantees equity for any given assignment, not instantly, but after some time.

But there are several questions:

- what is “equity” now
- what is “some time”
- why would drivers agree to give up control?
- what kind of system state emerges?

For Human drivers - idealistic **utopia**

For *Connected Autonomous Vehicles* - **practically applicable and desired(?)**



Formalization

Daily individual assignment

- Let's consider a demand Q on a single OD with K paths,
- some *assignment* (e.g. System Optimal) distributing it over K paths: Q_k^{SO} ,
- which yields $\hat{t}$ mean travel time ($\hat{t} < t^{UE!}$).



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- which yields $\hat{t}$ mean travel time ($\hat{t} < t^{UEI}$).

Such macroscopic assignment has its **daily realization** on day j , a binary matrix:

$$A_j \in \{0, 1\}^{Q \times K},$$

s.t.

$$\sum_{k=1}^K [A_j]_{i,k} = 1, \quad i = 1, \dots, Q,$$

$$\sum_{i=1}^Q [A_j]_{i,k} = Q_k, \quad k = 1, \dots, K.$$

Example: $Q = 5, K = 3$

$$A_j = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(Q_1, Q_2, Q_3) = (2, 2, 1)$$

Thus, every traveller is assigned to one path and the assignment is preserved.



Measuring inequity

For day j , individual deviations from the OD mean are

$$\mathbf{D}_j = \mathbf{A}_j \mathbf{T} - \hat{t}.$$

$\mathbf{D}_{j,i} > 0$: traveller i is worse off $\mathbf{D}_{j,i} < 0$: traveller i is better off



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After J days, the cumulative deviation is

$$\mathcal{D}_J = \sum_{j=1}^J \mathbf{D}_j.$$



Wardropian Cycles

A *Wardropian Cycle* of length n is a sequence of daily assignments $\mathbf{A} = (A_1, \dots, A_n)$ satisfying assignment on every day and $\mathcal{D}_n = \mathbf{0}$

after J days of following this sequence, all agents have experienced the same average travel time.

Theorem 1: (Existence of Wardropian Cycles)

For a given *OD* pair, for any route assignment with any fixed number of travelers, there *always* exists a finite Wardropian Cycle of length $\zeta = Q$.



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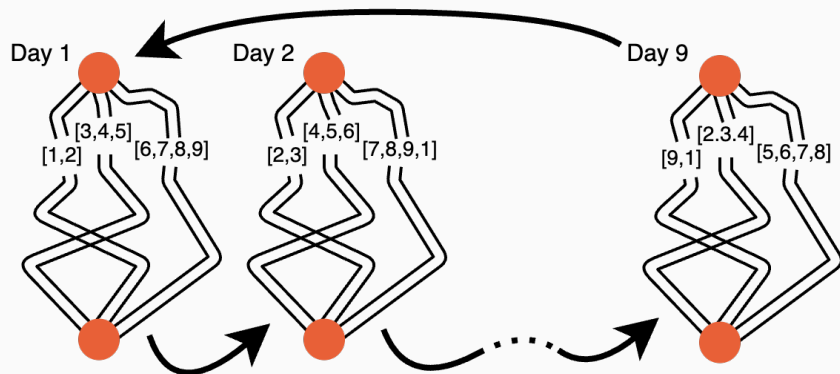
Theorem 1: (Existence of Wardropian Cycles)

For a given **OD** pair, for any route assignment with any fixed number of travelers, there **always** exists a finite Wardropian Cycle of length $\zeta = Q$.

For the Wardrop Equilibrium assignment, this theorem becomes trivially true with cycle length being one. **Is it general?**

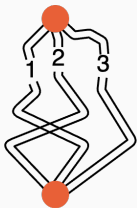


Example



Cycle length reduction - greatest common divisor

speed up the change of routes by GCD



Routes for this OD in SO		
route:	flow:	time:
1	20x 	15 min 
2	30x 	14 min 
3	40x 	9 min 

$$\hat{t} = 12 \text{ min}$$

Instead of a 90 day cycle, we can move agents in “batches” of 10, reducing the length of the cycle to just 9 days.

We called this method a GCD optimization.

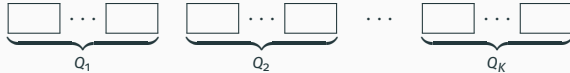


Cycle length reduction - greatest common divisor

The first reduction uses the **greatest common divisor** of the integer path counts:

$$M = \text{GCD}(Q_1, \dots, Q_K).$$

Assume that the rows of A_1 are arranged in route-specific blocks:



Since

$$M \mid Q_k \quad \forall k,$$

each route block consists of groups of M travellers.

$$A_{j+1} = \text{shift}_M(A_j)$$

GCD reduction

Shifting by M positions reduces length

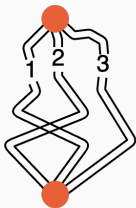
$$\zeta = Q/M.$$

Instead of shifting by one traveller, we shift by blocks of M :

$$1 \rightarrow 1 + M \rightarrow 1 + 2M \rightarrow \dots$$



Shortening the cycle: partition



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Observe, that any agent alternating between route 1 and 3 is in a Wardropian Cycle of length 2. Other agents can form a separate cycle of length 5.

Shortening the cycle: partition

Represent each path k by Q_k unit assignments with travel time t_k . Partition them into $J_1, \dots, J_S$ such that

$$\frac{1}{|J_s|} \sum_{j \in J_s} t_j = \hat{t}$$

$$s = 1, \dots, S.$$

Equal-mean subgroups

$$\underbrace{\{t_1, t_2, \dots\}}_{J_1} \quad \underbrace{\{t_4, t_5, \dots\}}_{J_2} \quad \dots$$

$$\text{mean}(J_1) = \text{mean}(J_2) = \dots = \hat{t}$$

A separate Wardropian Cycle is constructed within each subgroup:

$$\zeta_s = |J_s|.$$

Subgroups may have different cycle lengths and repeat independently.

Partition reduction

One cycle of Q travellers is replaced by smaller independent cycles:

$$J_1, \dots, J_S, \quad \sum_s |J_s| = Q.$$

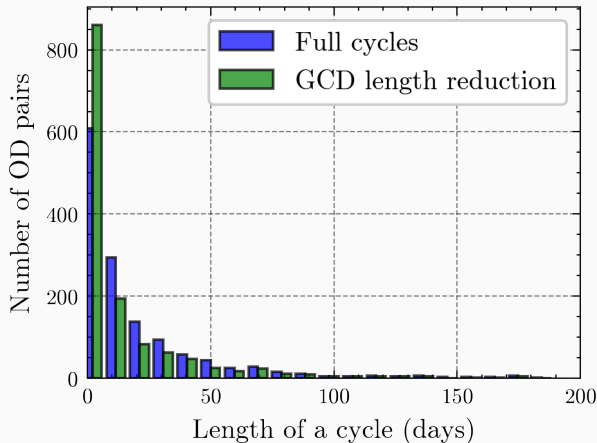
Computational difficulty

Finding a non-trivial equal-mean partition is NP-hard.



Applicability

While the Partition method is proven to be *NP-hard*, we applied first method to the real scale network:



Cycle from the perspective of an agent

We want agents to benefit from joining the Wardropian Cycle **as early as possible**.

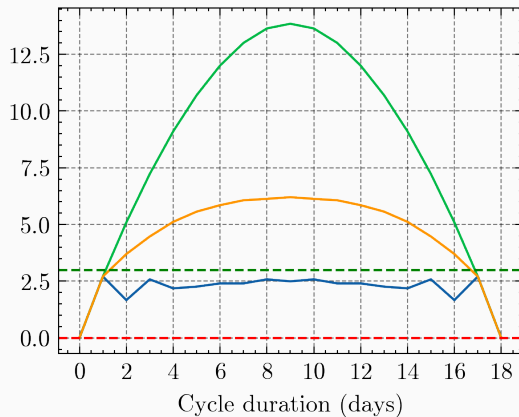
During the cycle, we therefore control the accumulated deviation $\mathbf{D}_j$ from the OD mean travel time $\hat{t}$:

$$\sum_{j=1}^n \mathbf{D}_j, \quad n = 1, \dots, \zeta.$$

Within-cycle inequity

The order of daily assignments can be chosen to bound the cumulative deviation experienced by each agent throughout the cycle.

Sum of absolute value of deviations from the average travel time (eq. 5)



A constructive algorithm follows directly from the bound



Cyclical User Equilibrium

For an OD pair, consider a cyclical assignment

$$\mathcal{A}^{WC} = (A_1, \dots, A_\zeta).$$

We call it a **Cyclical User Equilibrium (CUE)** if

$$\sum_{j=1}^{\zeta} \mathbf{D}_j = \mathbf{0}, \quad (\text{CUE-1})$$

$$\left[\sum_{j=1}^{\zeta} A_j \mathbf{T}_j \right]_i < t^{UE} \zeta, \quad \forall i \leq Q. \quad (\text{CUE-2})$$

CUE-1: fairness

$$\bar{t}_1 = \bar{t}_2 = \dots = \bar{t}_Q$$

No gain from exchanging positions within the cycle.

CUE-2: participation

$$\bar{t}_i^{WC} < t^{UE}$$

The cycle improves the long-run travel time relative to UE.



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If everyone participates, everyone is happy



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If everyone participates, everyone is happy

If anyone deviates - she/he is short term happy



Results

Barcelona case study

Network experiment

- 2 078 OD pairs
- static UE and SO assignments
- Wardropian Cycles constructed from SO path flows
- Greedy Equaliser applied over repeated days

UE $\rightarrow$ SO

$$T_{\text{UE}} = 1\,297\,794 \text{ min}$$

$$T_{\text{SO}} = 1\,268\,541 \text{ min}$$

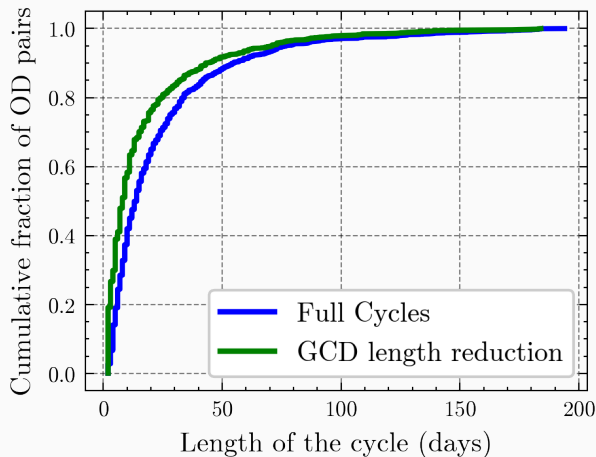
$$\text{PoA} \simeq 1.02$$

$$\Delta T = 29\,253 \text{ min} \quad (\approx 2.3\%)$$

Exact cycles exist for the analysed OD pairs, but can be long: median cycle length 16 days (11 after GCD reduction).



Barcelona: exact cycles can be long



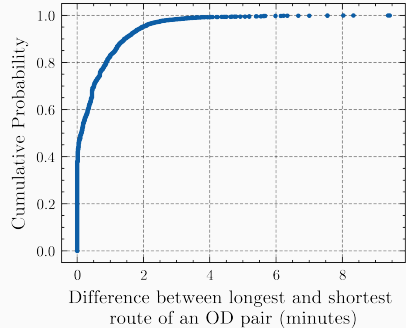
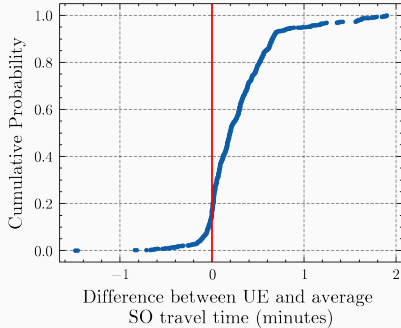
Cycle	Mean	Median	95th
Full	31.4	16	100.3
GCD	25.9	11	91

The longest cycle is 777 days under both constructions.

Exact periodic schedules therefore become impractical for part of the demand, which motivates the Greedy Equaliser.



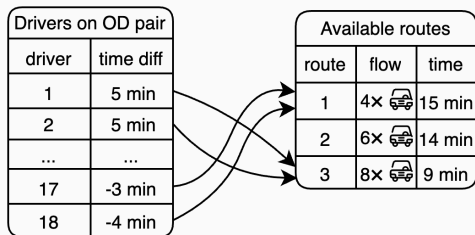
Barcelona: efficiency and distribution across OD pairs



- The aggregate gain is not uniform: **17.5%** of OD pairs have a higher mean travel time under the city-wide SO assignment than under UE.
- For about **80%** of OD pairs, the difference between the shortest and longest used SO path is below one minute.



Generalizations – the greedy algorithm



- Keep the prescribed route counts for the current day.
- Rank travellers by accumulated travel-time disadvantage.
- Rank available route positions by current travel time.
- Match the fastest positions to the most disadvantaged travellers.

This rule can be applied repeatedly without constructing an exact cycle in advance.



Algorithm 1 Greedy Equaliser

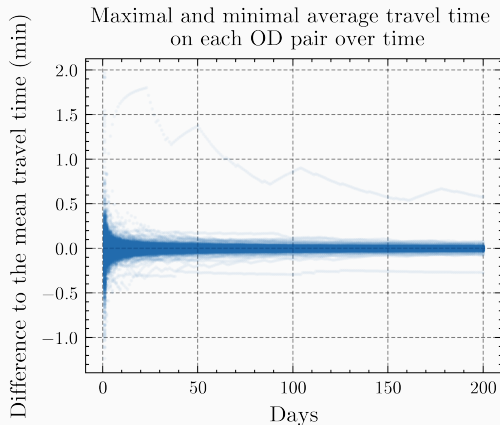
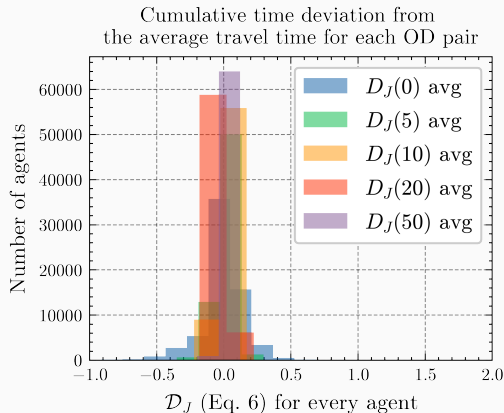
Require: Q^j {daily demand (participating drivers)}
Require: $Q_k(Q^j)$ {path assignment (externally computed)}
Require: $\bar{l}_{j-1}$ {cumulated inequity of participating drivers}
 $\pi_j(\bar{l}_j) \leftarrow$ drivers sorted in descending order of inequity $\bar{l}_{j-1}$
 $P \leftarrow$ paths sorted k in ascending order of travel time t_k
 $k \leftarrow 0$ {index of currently loaded path}
 $\forall k \quad \hat{Q}_k \leftarrow 0$
for each driver i in π_j **do**
 $[A_j]_{i,P[k]} = 1$ {assign driver i to path $P[k]$ }
 $\hat{Q}_{P[k]} \leftarrow \hat{Q}_{P[k]} + 1$ {update path flow}
 if $\hat{Q}_{P[k]} = Q_{P[k]}$ **then**
 $k \leftarrow k + 1$ {move to the next fastest path}
 end if
end for
 $t_j \leftarrow \text{SIM}(A_j)$ {realised travel times}
 $\bar{l}_j \leftarrow \bar{l}_j(t_j)$ {update inequity using realised travel times}
Ensure: $\bar{l}_j$ {updated inequity used on the next day}

Proposition

For fixed route counts and route travel times on day $J + 1$, the **Greedy Equaliser minimises the next-day quadratic inequity** $\bar{l}_{J+1}$ among all feasible assignments preserving those route counts.



Barcelona: Greedy Equaliser



Repeated greedy assignments narrow the distribution of cumulative deviations and reduce the gap between the most and least disadvantaged travellers within OD pairs.



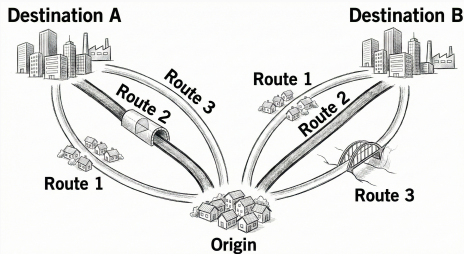
Replication across networks

Network	PoA	l_5/l_1	l_{10}/l_1	l_{20}/l_1	l_{50}/l_1
Barcelona	1.02	0.16	0.07	0.06	0.04
Anaheim	1.01	0.15	0.05	0.02	0.01
Eastern Massachusetts	1.03	0.17	0.16	0.13	0.06
Berlin	1.03	0.10	0.03	0.01	0.01
Sioux Falls	1.04	0.11	0.04	0.02	0.01

After five assignments, less than **20%** of the initial inequity remains in all five networks; convergence rates differ with the route-flow and route-time structure.

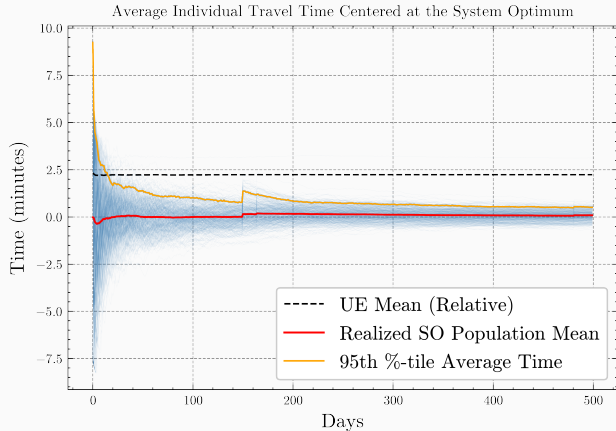


Synthetic example: varying participation and stochastic travel times



- 1000 travellers repeatedly choose between destination A, destination B, and staying home.
- Daily participation therefore changes; travellers to each destination are assigned among three routes.
- Realised travel times are stochastic and are not perfectly predicted by the controller.
- A rare disruption increases travel time on a route; subsequent assignments use realised accumulated disadvantage.

Synthetic example: Greedy Equaliser response



Under the simulated conditions, individual mean travel times move towards the coordinated-assignment mean. The disruption around day 150 creates a temporary loss that is subsequently compensated through the accumulated-disadvantage rule.



Limitations and applicability

- **Participation and compliance.** Wardropian Cycles apply to travellers who participate in the prescribed daily assignment. This setting is most direct for centrally managed fleets, mobility platforms, or opt-in participant pools.
- **Partial control.** With partial control, the prescribed route split must account for the response of self-interested traffic. Once a feasible split is obtained, the Greedy Equaliser determines which participating user receives each route position.
- **Dynamic conditions.** Exact cycles assume fixed daily route flows and path times. The Greedy Equaliser can update accumulated deviations from realised daily outcomes, while the traffic dynamics generating those outcomes remain external to the assignment rule. - Karma poster by **Anastasia Psarou + ETH** yesterday
- **Heterogeneous users.** Equal travel-time deviations need not imply equal experienced utility when travellers differ in value of time, schedule constraints, route preferences, or flexibility.



Aggregate assignment determines the route-flow target.

Temporal assignment determines how repeated users share the faster and slower route positions generated by that target.

Wardropian Cycles can make traffic routing
for compliant connected autonomous vehicles (**CAVs**)
both optimal and fair.



Thank you

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- Michał Hoffman (now at Oxford)
- Michał Bujak (now at EPFL)
- Grzegorz Jamróz (finished PD yesterday)
- Rafał Kucharski

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Thank you. Any questions?

